Quantum Teleportation Algorithm

Quantum AI (417523A) - SPPU BE AI&DS

CODE

```
import math
import warnings

from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister, transpile
from qiskit_aer import AerSimulator

warnings.filterwarnings("ignore", category=DeprecationWarning)

SHOTS = 2048

print("=" * 66)
print("  QUANTUM AI (417523A)  -  PRACTICAL 2")
print("  QUANTUM TELEPORTATION ALGORITHM")
print("=" * 66)

q = QuantumRegister(3, "q")
c = ClassicalRegister(3, "c")
qc = QuantumCircuit(q, c)

qc.ry(1.234, q[0])
print("\nAlice wants to send the unknown state RY(1.234)|0> on q0 to Bob.")

qc.h(q[1])
qc.cx(q[1], q[2])
print("Entangled pair (Bell state) created between q1 and q2.")

qc.cx(q[0], q[1])
qc.h(q[0])
qc.barrier()

qc.measure(q[0], c[0])
qc.measure(q[1], c[1])
print("Bell-basis measurement performed on q0 and q1; two classical")
print("bits are sent to Bob over an ordinary channel.")

with qc.if_test((c[1], 1)):
    qc.x(q[2])
with qc.if_test((c[0], 1)):
    qc.z(q[2])

qc.barrier()
qc.measure(q[2], c[2])

sim = AerSimulator()
counts = sim.run(transpile(qc, sim), shots=SHOTS).result().get_counts()

print(f"\nMeasurement results over {SHOTS} shots  (bit order: c2 c1 c0):")
for state, count in sorted(counts.items(), key=lambda kv: -kv[1]):
    print(f"   |{state}> : {count:5d}   ({100 * count / SHOTS:5.2f} %)")

recovered = sum(v for k, v in counts.items() if k[0] == "1")
print("\nClassical bits c1 c0 are perfectly random, and the conditional")
print("X / Z feed-forward leaves qubit 2 in the original state.")
print(f"Probability of measuring |1> on qubit 2 : {recovered / SHOTS:.4f}")
print(f"Expected for RY(1.234)|0>               : {(1 - math.cos(1.234)) / 2:.4f}")
print("Qubit 2 therefore recovers Alice's original state exactly.")
print("=" * 66)
```

OUTPUT

```
==================================================================
  QUANTUM AI (417523A)  -  PRACTICAL 2
  QUANTUM TELEPORTATION ALGORITHM
==================================================================
```

```
Alice wants to send the unknown state RY(1.234)|0> on q0 to Bob.
Entangled pair (Bell state) created between q1 and q2.
Bell-basis measurement performed on q0 and q1; two classical
bits are sent to Bob over an ordinary channel.

Measurement results over 2048 shots  (bit order: c2 c1 c0):
   |000> :   356   (17.38 %)
   |011> :   344   (16.80 %)
   |001> :   338   (16.50 %)
   |010> :   317   (15.48 %)
   |110> :   177   ( 8.64 %)
   |111> :   176   ( 8.59 %)
   |100> :   174   ( 8.50 %)
   |101> :   166   ( 8.11 %)

Classical bits c1 c0 are perfectly random, and the conditional
X / Z feed-forward leaves qubit 2 in the original state.
Probability of measuring |1> on qubit 2 : 0.3384
Expected for RY(1.234)|0>               : 0.3348
Qubit 2 therefore recovers Alice's original state exactly.
======================================================================
```
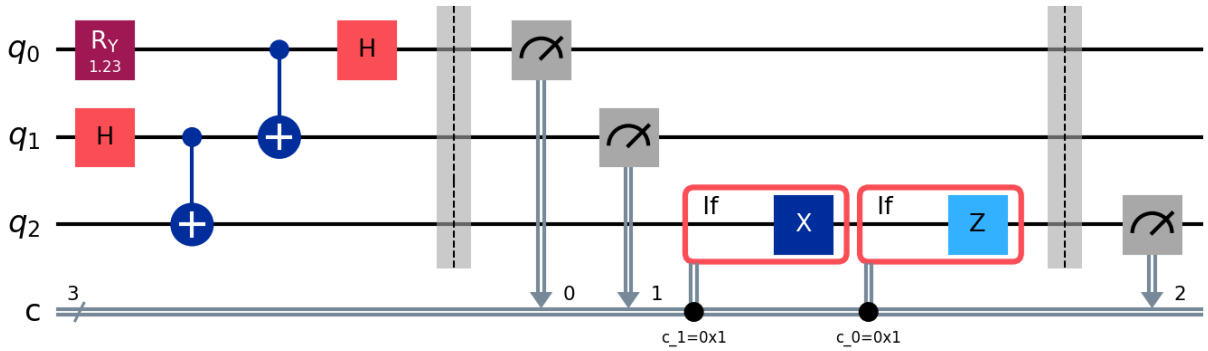
FIGURE: Q2_teleportation_circuit